

9th chapter
(Short question)
of PP 2014-23

{2023}

(5) On page no. 274



(2) Dirty Dozen refers to twelve persistent organic pollutants identified by the UN as highly hazardous due to their longevity, bioaccumulation and adverse health effects. The list includes:

- Aldrin
- chlordane
- DDT
- Dieldrin
- Endrin
- Heptachlor
- Hexachlorobenzene
- mirex
- PCBs
- Toxaphene
- Dioxins
- Furans



(3) PCBs are synthetic organic compounds/chemicals used in industrial applications such as coolants, insulating fluids, plasticizers and lubricants due to their stability and non-flammability.

They are highly toxic, persistent in environment and bioaccumulate in the food chain. PCBs can cause cancer, immune system suppression, reproduction disorders, and endocrine disruption. Their production and use were banned or restricted in the late 1970s.

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{2022}

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(13)

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(5) On page no. ~~200~~ 273

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{2021}

(2) There are following impact of chemicals on buildings:

- (i) Corrosion: Acid rain corrodes limestone, marble and steel. This weakens structural components and can lead to costly repairs overtime.
- (ii) Material degradation: chlorides and sulfates weaken bricks and mortar. continuous exposure accelerates decay, compromising building stability.
- (iii) Paint Damage: Pollutants cause paint to peel and fade. This not only

affects the building's appearance but also reduces its protective layer.



(2) On page no. 274



{2020}

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(4)



(9) PCBs were banned due to their high toxicity, persistence in the environment, bioaccumulative nature and associated health risks, including cancer and reproductive disorders.



{2019}

(d) PCBs are harmful to the environment due to their persistence, bioaccumulation and toxicity. They resist degradation, accumulate in organisms and become more concentrated up the food chain, affecting wildlife and human health. PCBs contaminate soil, water and air, leading to the long-term environmental issues.



{2018}

(4) The impacts of chemicals on visibility are:

(i) Smog formation:

Chemical pollutants react to form smog, reducing urban visibility.

(ii) Particulate matter:

Fine particles from combustion sources scatter light, causing haze and reduced clarity.

(iii) Acid Deposition:

Chemicals contribute to acid rain, scattering light and decreasing visibility.

(iv) Ozone:

Ground level ozone, formed from pollutants in sunlight, reduces visibility in urban areas.

(5) on page no. 264.

(6) The impacts of silver on human beings are:

(i) Antimicrobial Benefits:

Silver is used for its antimicrobial

properties in medical settings.

(ii) Allygia Risk:

Excessive exposure can cause permanent skin discoloration (argyria).

(iii) Industrial exposure:

Workers may face respiratory issues or skin irritation.

(iv) Dietary caution:

Supplements containing silver should be used cautiously to avoid health risks.



{2017}

(7) On page no. 264 & 267



(9) Chemicals impact monuments and buildings by causing:

- Corrosion:

Acid rain corrodes limestone, marble and metals.

- Discoloration:

Pollutants stain and discolor building surfaces.

- Weathering:

Chemicals accelerate natural stone and concrete degradation.

- Efflorescence:

Soluble salts leave white deposits on masonry.

- Biological growth:

Some chemicals promote algae and fungi growth, damaging surfaces.

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[2016]

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(6) On Page no. 279

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[2015]

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(6) The detrimental effects of pesticides in aqueous environment are:

- Toxicity:

Pesticide can be toxic to aquatic life, including fish and invertebrates.

- Bioaccumulation:

They can accumulate in aquatic organisms, posing risks to predators and humans.

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- Ecosystem Disruption:

Pesticide disrupt aquatic food chains and habitats.

- Water Quantity:

Runoff contaminates water bodies, harming aquatic ecosystems and water quality.



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(5), (6), (9)

